

Instantons

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ABSTRACT: These are notes based on Sydney Coleman's lectures titled "*Uses of Instantons*". I am writing this primarily because there are points in the book that are too quick for me, and I need more thought to understand it clearly. I had great discussions with Prof. Peter Arnold, which helped me understand the material a lot better.

Contents

1	Instantons in Quantum Mechanics	1
1.1	The Big Picture	1
1.2	The Strategy	3
1.3	Semi-Classical Approximation	5
1.4	The Harmonic Oscillator	8
1.5	Instantons	10
1.6	The Instanton Gas	16
1.7	Periodic Potentials and Bloch Waves	19
1.8	Unstable States and Bounces	21

1 Instantons in Quantum Mechanics

The organization of this section is as follows. We start with the big picture in section 1.1, explaining the phenomena that we are trying to describe. Then, we will describe the approach that we are going to take in section 1.2. A general result for all the later calculations is derived in section 1.3. We then turn to calculations for the tunneling phenomena in the next sections 1.4 and 1.5, and finally introducing instantons in 1.5. In section 1.6, we will learn about the instanton dilute gas and how it gives the correct results for tunneling. We will test our understanding of the instanton technique by applying it to two different cases in sections 1.7 and 1.8.

1.1 The Big Picture

We will start by painting the big picture. In our undergraduate classes, we learn perturbation theory to handle interactions that are typically weak. And indeed, we feel that when the interactions are small, perturbation theory is all there is. But, perturbation theory fails to capture one very important class of phenomena in quantum mechanics: *tunneling*.

Let us take a simple system, and understand why tunneling is not part of the standard perturbation theory. Consider a particle moving in a potential, given by the Lagrangian,

$$L(x, \dot{x}) = \frac{1}{2} \dot{x}^2 - V(x; g) \tag{1.1}$$

Here, g is called the *coupling*. We will take the potential to be the form,

$$V(x; g) = \frac{1}{g^2} f(gx)$$

and assume that $f(x)$ starts with x^2 . The reason for this choice of form is motivated from our ultimate interest in quantum field theory. If the potential is of this form, then classically the coupling g is not important at all. To see this, all we have to do is a re-scaling of x ,

$$x' = gx$$

Then, the Lagrangian becomes,

$$L = \frac{1}{g^2} \left(\frac{1}{2} \dot{x}'^2 - V(x'; g = 1) \right)$$

Clearly, the equations of motion will not involve g at all. In hindsight, this can be seen without doing any of the re-scaling. Since g is dimensionful, one can set g to unity with appropriate choice of units. The same is true for Yang-Mills theory. The Yang-Mills coupling is classically unimportant.

Things are different in quantum mechanics. There is a new dimensionful constant, $\hbar$, which changes the story. The equation of motion now takes a less important role in the game, and the central player becomes,

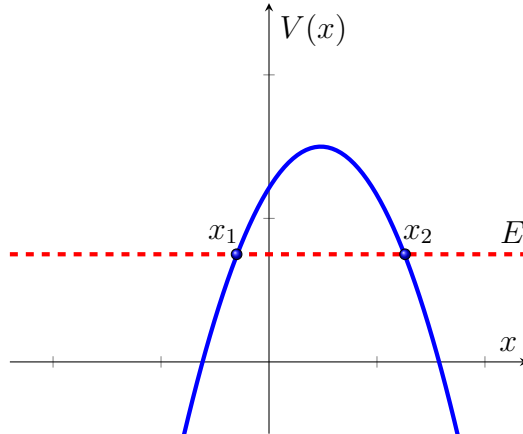
$$\exp \left\{ -\frac{1}{g^2 \hbar} \int d\tau L \right\}$$

Now, $g^2 \hbar$ is the dimensionless parameter that is important. The semi-classical approximations ($\hbar \rightarrow 0$) are now equivalent to small coupling ($g \rightarrow 0$) approximations.

We are now in a position to understand why perturbation theory is inadequate to explain tunneling. A perturbative series is a power series in g or equivalently $\hbar$. But we know that the probability of barrier penetration is given by,

$$|T(E)| = \exp \left\{ -\frac{1}{\hbar} \int_{x_1}^{x_2} dx \sqrt{2(V(x) - E)} \right\} (1 + \mathcal{O}(\hbar)) \quad (1.2)$$

where E is the energy of the particle, and x_1 and x_2 are the classical turning points, as shown in the figure,



The tunneling amplitude cannot be written as a power series in $\hbar$. Clearly, it vanishes faster than any power of $\hbar$. Therefore, perturbation theory will be blind to these phenomena!

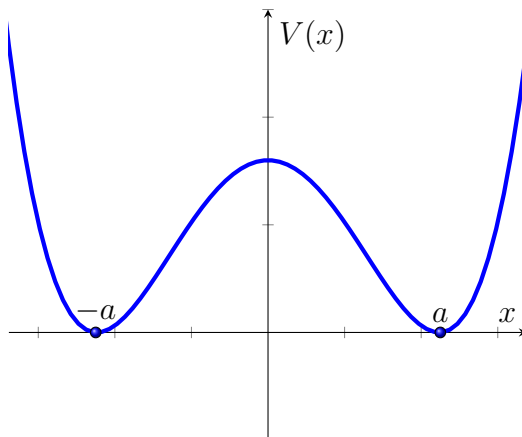
Although we have a nice approximation technique to describe tunneling, – namely the WKB method – it does not generalize to quantum field theory easily. Therefore, we will reformulate tunneling in a new way, which can be easily extended to quantum fields. This is the technique of instantons.

1.2 The Strategy

To have a clear idea of what we are looking for, let us first setup the problem. Consider the double well potential,

$$V(x) = \frac{\omega^2}{8a^2}(x^2 - a^2)^2 \quad (1.3)$$

A plot of the potential looks like,



From this picture, we expect the system to have two ground states, localized at each of the minima $x = \pm a$. But it turns out that the system will have one unique ground state, which is the symmetric combination of the “would have been ground states”. The antisymmetric combination – which now is the first excited state – has a very small gap from the ground state. The reason – you guessed it! – is tunneling. Our goal is to calculate this splitting energy. As mentioned before, this splitting energy do not appear as a result of perturbations breaking the degeneracy of the “would have been ground states”. This splitting has a “non - perturbative” origin.

Our goal is to re-phrase this question in terms of the path integral. First, let us collect some important facts about path integrals. For the Hamiltonian given by,

$$H = \frac{p^2}{2} + V(x) \quad (1.4)$$

The transition amplitude from any position x_i to any other position x_f in time T is,

$$\langle x_f | e^{-\hat{H}T/\hbar} | x_i \rangle = \int_{x(-T/2)=x_i}^{x(T/2)=x_f} \mathcal{D}x(\tau) \exp \left\{ -\frac{1}{\hbar} \int_{-T/2}^{T/2} d\tau \left(\frac{1}{2} \dot{x}^2 + V(x) \right) \right\} \quad (1.5)$$

We are working in imaginary (or Euclidean) time. What sits in the exponential on the right-hand side is the Euclidean action corresponding to the Hamiltonian in (1.4),

$$S[x(\tau)] = \int_{-T/2}^{T/2} d\tau \left(\frac{1}{2} \dot{x}^2 + V(x) \right) \quad (1.6)$$

A very important feature of the Euclidean path integral that we will utilize copiously is the following,

“If we wait long enough, things evolve to the lowest energy state.”

Let us unpack this slogan. Let $\{|n\rangle\}$ be the eigen states of the Hamiltonian, $\hat{H}$. Then, we have,

$$|x_i\rangle = \sum_n |n\rangle \psi_n^*(x_i) \quad \text{and} \quad \langle x_f| = \sum_n \langle n| \psi_n(x_f)$$

where, $\psi_n(x) = \langle x|n\rangle$ are the energy eigen wavefunctions. Using this on the left-hand side of (1.5), we get,

$$\langle x_f | e^{-\hat{H}T/\hbar} | x_i \rangle = \sum_n e^{-E_n T/\hbar} \psi_n(x_f) \psi_n^*(x_i) \quad (1.7)$$

If we wait long enough, all the high energy states will be suppressed. In particular, if x_i and x_f has non-zero overlap with the ground state, we will have,

$$\langle x_f | e^{-\hat{H}T/\hbar} | x_i \rangle \xrightarrow{T \rightarrow \infty} e^{-E_0 T/\hbar} \psi_0(x_f) \psi_0^*(x_i) \quad (1.8)$$

Therefore, to extract the ground state energy (or wavefunction), we simply have to find the infinite time behavior of some transition amplitude, assuming the initial and final states do have a non-zero overlap with the ground state.

Now, let us once again recall our aim. We want the splitting energy between the ground state and the first excited states of the double well potential. From symmetry alone, we can guess the qualitative features of the ground state and first excited state. Classically, as we said before, we expect two ground states, each centered at $\pm a$. We can then say with confidence that $|\pm a\rangle$ will have non-zero overlaps with these “would have been ground states” centered at the respective positions. Since the potential has parity symmetry, the ground state (and all other energy eigen states) should be either odd or even. We know that the ground state have no nodes. Therefore, our first guess for the ground state will be the symmetric combination for the “would have

been ground states”, and first excited state is the antisymmetric combination. So, it is a reasonable guess that the true ground state will have non-zero overlap with this symmetric combination. i.e, if $|0\rangle$ is the true ground state, then we expect,

$$\langle 0 | (|a\rangle + |-a\rangle) \neq 0 \quad (1.9)$$

Similarly, the true first excited state, $|1\rangle$ will have non-zero overlap with the anti-symmetric combination,

$$\langle 1 | (|a\rangle - |-a\rangle) \neq 0 \quad (1.10)$$

Let us give names,

$$|\text{ground}\rangle = |a\rangle + |-a\rangle \quad \text{and} \quad |\text{first}\rangle = |a\rangle - |-a\rangle$$

Then, we can confidently say the following,

$$\langle \text{ground} | e^{-\hat{H}T/\hbar} | \text{ground} \rangle \xrightarrow{T \rightarrow \infty} e^{-E_0 T/\hbar} (\text{stuff}) \quad (1.11a)$$

and,

$$\langle \text{first} | e^{-\hat{H}T/\hbar} | \text{first} \rangle \xrightarrow{T \rightarrow \infty} e^{-E_1 T/\hbar} (\text{stuff}) \quad (1.11b)$$

For the second of the above to be true, we need $|\text{first}\rangle$ to have zero overlap with the ground state. This is expected to be true because the ground state is symmetric and $|\text{first}\rangle$ is antisymmetric.

Therefore, to find the splitting energy, the computations we have to do have been boiled down to two transition amplitudes,

$$\langle a | e^{-\hat{H}T/\hbar} | a \rangle = \langle -a | e^{-\hat{H}T/\hbar} | -a \rangle \quad \text{and} \quad \langle a | e^{-\hat{H}T/\hbar} | -a \rangle = \langle -a | e^{-\hat{H}T/\hbar} | a \rangle$$

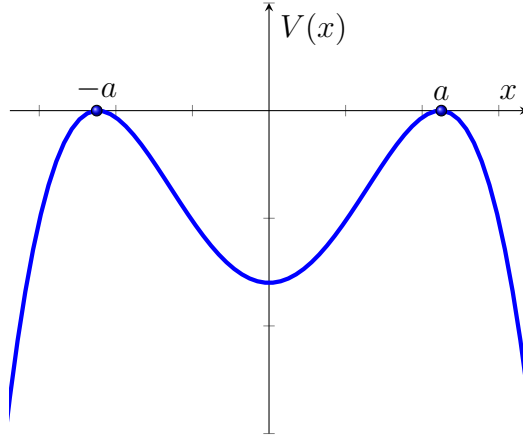
The equalities are again due to the symmetry of the potential. We will calculate these in the semi-classical approximation in section 1.6.

1.3 Semi-Classical Approximation

We will take some time to set up the calculation in general, before plunging into details. Of course, we cannot calculate the path integrals exactly. What we are interested in is the semi-classical approximation, or the small $\hbar$ approximation. We know that as we take $\hbar \rightarrow 0$, the path integral will get its major contribution from the “classical” path, which is the path that minimizes the action. Let us denote this path by $x_{\text{cl}}(\tau)$. The classical path will obey the equations of motion,

$$-\frac{d^2 x_{\text{cl}}(\tau)}{d\tau^2} + V'(x_{\text{cl}}) = 0 \quad (1.12)$$

and satisfy the required boundary conditions. If we want to import our mechanics intuition, we can think of it as a particle moving in a potential *minus* $V(x)$. That is, for the double well potential in (1.3), the corresponding potential will look like,



Now comes the key step of our semi-classical approximation. We will write any path $x(\tau)$ as,

$$x(\tau) = x_{\text{cl}}(\tau) + \delta x(\tau) \quad (1.13)$$

with the assumption that $\delta x(\tau)$ are all small fluctuations. The fluctuations vanish at the boundary (i.e, $\delta(\pm T/2) = 0$), because $x_{\text{cl}}(\tau)$ already takes care of the boundary conditions. We will then expand the action around the classical path,

$$S[x_{\text{cl}} + \delta x] = S[x_{\text{cl}}] + \int d\tau \frac{1}{2} \delta x(\tau) \left(-\frac{d^2}{d\tau^2} + V''(x_{\text{cl}}) \right) \delta x(\tau) + \dots$$

The term linear in δx will vanish due to the equation of motion. We will ignore all the terms higher than quadratic in δx , and we will have an approximate path integral,

$$\langle x_f | e^{-\hat{H}T/\hbar} | x_i \rangle \approx e^{-S_{\text{cl}}/\hbar} \int \mathcal{D}\delta x(\tau) \exp \left\{ -\frac{1}{\hbar} \int_{-T/2}^{T/2} d\tau \frac{1}{2} \delta x(\tau) \left(-\frac{d^2}{d\tau^2} + V''(x_{\text{cl}}) \right) \delta x(\tau) \right\} \quad (1.14)$$

An Aside : Saddle Point Approximation

Let me go through the saddle point approximation in calculus briefly. We consider an integral of the type,

$$I = \int_{\mathbb{R}} dx e^{-S(x)/\hbar}$$

Assume that $S(x)$ has a minimum, at x_0 . Then, for small $\hbar$, the majority of the contribution to the integral will come from $S(x_0)$. To get the leading correction, we can simply expand S around the minimum,

$$S[x] = S[x_0] + \frac{1}{2}(x - x_0)^2 S''(x_0) + \dots$$

We will ignore all the terms after quadratic. In general, the expansion only makes sense around x_0 or when $\delta x = x - x_0$ is small. We can therefore approximate the

integral as,

$$I \approx e^{-S_0/\hbar} \int_{-\epsilon}^{\epsilon} d(\delta x) e^{-\frac{1}{2}\delta x^2 S''(x_0)/\hbar}$$

But for small $\hbar$, the integral over δx is dominated by $\delta x = 0$. Therefore, we can extend the integration region from $(-\epsilon, \epsilon) \rightarrow \mathbb{R}$ with very little cost. Then, we have the approximate answer to be,

$$I \approx \sqrt{\frac{2\pi\hbar}{S''_0}} e^{-S_0/\hbar}$$

Back to Semi-Classical Approximation

We now have a better understanding of the approximation (1.14). The path integral over $\delta x(\tau)$ is over all possible variations, not just small variations, and this is okay.

To evaluate the integral in (1.14), we do the following. We write δx as a sum of orthonormal basis,

$$\delta x(\tau) = \sum_{n=0}^{\infty} c_n x_n(\tau) \quad (1.15)$$

By orthonormal, what we mean is obviously,

$$\int_{-T/2}^{T/2} d\tau x_m(\tau) x_n(\tau) = \delta_{mn} \quad (1.16)$$

Also, these basis functions should vanish at the end points. Staring at (1.14), it is natural to choose these basis functions to be the eigen states of the operator in the exponent. That is, we choose x_n such that,

$$\left(-\frac{d^2}{d\tau^2} + V''(x_{cl}) \right) x_n(\tau) = \lambda_n x_n(\tau) \quad (1.17)$$

We will assume that all the eigen values are non-zero and positive. This assumption will soon turn out to be false. We will deal with it then. Now, we then will end up with,

$$\begin{aligned} \langle x_f | e^{-\hat{H}T/\hbar} | x_i \rangle &\approx e^{-S_{cl}/\hbar} \int_{\mathbb{R}} \prod_{n=0}^{\infty} \frac{dc_n}{\sqrt{2\pi\hbar}} \exp \left\{ -\frac{1}{\hbar} \prod_{n=0}^{\infty} \frac{\lambda_n c_n^2}{2} \right\} \\ &= e^{-S_{cl}/\hbar} \prod_{n=0}^{\infty} \frac{1}{\sqrt{\lambda_n}} \\ &= e^{-S_{cl}/\hbar} \frac{1}{\sqrt{\det\{-\partial_\tau^2 + V''(x_{cl})\}}} \end{aligned} \quad (1.18)$$

The above is an approximate formula and there are corrections of the order $\hbar$. But we will not bother writing “ $+\mathcal{O}(\hbar)$ ” every time.

1.4 The Harmonic Oscillator

Now, we will roll up our sleeves and start calculating. We will first compute,

$$\langle a | e^{-\hat{H}T/\hbar} | a \rangle = \langle -a | e^{-\hat{H}T/\hbar} | -a \rangle$$

The initial and final positions of the transition amplitude is one of the minima. The classical solution is thus a particle starting on top of an upside down hill, and ending up at the same place, after some time T (which we will take to ∞ later.).

It seems like the only solution we have is $x_{\text{cl}}(\tau) = a$ (or $-a$. We will just do the problem for a). We will later see that there are approximate solutions other than this trivial one due to instantons. For now, let us work with this one true classical solution.

Since the particle sits on top of the hill at every moment, we have $S_{\text{cl}} = 0$. From our potential (1.3), we can compute,

$$V''(x_{\text{cl}}) = V''(a) = \omega^2 \quad (1.19)$$

We can therefore just use our general result (1.18). We just need to find,

$$\frac{1}{\sqrt{\det\{-\partial_\tau^2 + \omega^2\}}}$$

First, some good news. The eigen values are all positive. We can understand this by thinking of the operator $D = -\partial_\tau^2 + \omega^2$ as a massive Klein-Gordon operator. Since it is massive, the smallest eigen value is the “mass”, which is $\omega \neq 0$. From the analogy, we can guess that the eigen functions are some oscillatory functions. But remember, we should ensure that the eigen functions should vanish at $\pm T/2$. Therefore, we have the eigen value equation,

$$(-\partial_\tau^2 + \omega^2) \sin\left(\frac{2\pi n\tau}{T}\right) = \lambda_n \sin\left(\frac{2\pi n\tau}{T}\right) \quad (1.20)$$

We have,

$$\lambda_n = \omega^2 + \left(\frac{2\pi n}{T}\right)^2 \quad (1.21)$$

The eigen values are all positive as promised. Then, we can find what we need by,

$$\frac{1}{\sqrt{\det\{-\partial_\tau^2 + \omega^2\}}} = \prod_{n=-\infty}^{\infty} \frac{1}{\sqrt{\omega^2 + \left(\frac{2\pi n}{T}\right)^2}}$$

Using the $n \rightarrow -n$ symmetry of the eigenvalues, we can write,

$$\frac{1}{\sqrt{\det\{-\partial_\tau^2 + \omega^2\}}} = \frac{1}{\omega} \prod_{n=1}^{\infty} \frac{1}{\omega^2 + \left(\frac{2\pi n}{T}\right)^2} \quad (1.22)$$

Zeta Function Regularization

We will evaluate this product via a technique called the zeta function regularization. We first make some reshuffling,

$$\frac{1}{\omega} \prod_{n=1}^{\infty} \frac{1}{\omega^2 + \left(\frac{2\pi n}{T}\right)^2} = \frac{1}{\omega} \prod_{m=1}^{\infty} \left(\frac{2\pi m}{T}\right)^{-2} \prod_{n=1}^{\infty} \left[1 + \left(\frac{T\omega}{2\pi n}\right)^2\right]^{-1}$$

The second product is a well known result (I do not know the proof myself, I am just quoting the result),

$$\prod_{n=1}^{\infty} \left[1 + \left(\frac{T\omega}{2\pi n}\right)^2\right] = \frac{2}{T\omega} \sinh \frac{T\omega}{2}$$

For the first product, we use the zeta function technique. The zeta function is defined as,

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

Therefore,

$$\left(\frac{T}{2\pi}\right)^{2s} \zeta(2s) = \sum_{n=1}^{\infty} \left(\frac{2\pi n}{T}\right)^{-2s}$$

Differentiate with respect to s ,

$$2\left(\frac{T}{2\pi}\right)^{2s} \ln\left(\frac{T}{2\pi}\right) \zeta(2s) + 2\left(\frac{T}{2\pi}\right)^{2s} \zeta'(2s) = \sum_{n=1}^{\infty} (-2) \left(\frac{2\pi n}{T}\right)^{-2s} \ln\left(\frac{2\pi n}{T}\right)$$

The motive to differentiate comes from the log term that sits inside the sum in the right-hand side. If we set $s = 0$,

$$2 \ln\left(\frac{T}{2\pi}\right) \zeta(0) + 2\zeta'(0) = \ln\left\{\prod_{n=1}^{\infty} \left(\frac{2\pi n}{T}\right)^{-2}\right\}$$

Using $\zeta(0) = -\frac{1}{2}$ and $\zeta'(0) = -\frac{1}{2} \ln 2\pi$, we have,

$$\ln\left\{\prod_{n=1}^{\infty} \left(\frac{2\pi n}{T}\right)^{-2}\right\} = -\ln T \implies \prod_{n=1}^{\infty} \left(\frac{2\pi n}{T}\right)^{-2} = \frac{1}{T} \quad (1.23)$$

We can plug these results to (1.22), and get,

$$\langle a | e^{-\hat{H}T/\hbar} | a \rangle \approx \frac{1}{\sqrt{\det\{-\partial_\tau^2 + \omega^2\}}} = \frac{1}{2 \sinh \frac{T\omega}{2}} \quad (1.24)$$

The large time behavior is,

$$\langle a | e^{-\hat{H}T/\hbar} | a \rangle \approx e^{-\omega T/2} \quad (1.25)$$

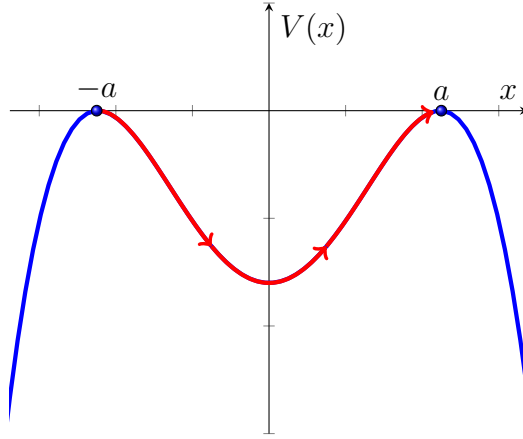
The energy of the state localized at a is $E_0 = \frac{\hbar\omega}{2}$. This is the correct semi-classical result. For small $\hbar$, the particle will be in a harmonic oscillator ground state centered around the minimum.

1.5 Instantons

We finally come to the main character of these notes, instantons. We will calculate the transition amplitude,

$$\langle a | e^{-\hat{H}T/\hbar} | -a \rangle$$

using semi-classical approximation. First, we should find the classical path, which starts at $-a$ and ends at $+a$ in time T on the upside down potential. In pictures, the path will look like the red curve below,



In finite time T , there are no such paths. But we are ultimately interested in taking $T \rightarrow \infty$, and such a path exists in this case!

We can use energy conservation to solve for the path (or equations of motion. This way is easier). We have a conserved energy given by,

$$E = \frac{1}{2}\dot{x}^2 - V(x) \quad (1.26)$$

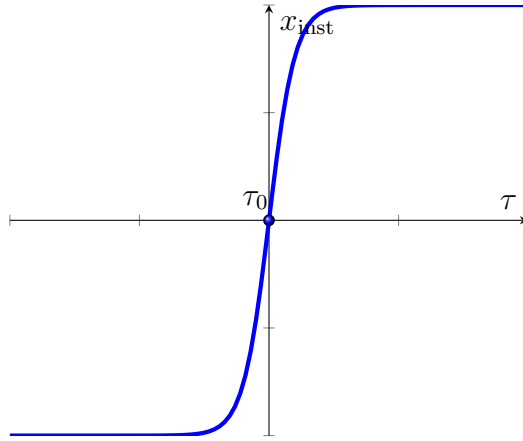
If the particle is starting from the top of the hill with zero velocity, then the energy there is zero. Therefore, we have the solution,

$$\tau = \tau_0 + \int_0^{x_{\text{inst}}} dx \frac{1}{\sqrt{V(x)}} \quad (1.27)$$

τ_0 is an integration constant, and denotes the time at which the path crosses the origin (i.e, $x = 0$ when $\tau = \tau_0$). This solution $-x_{\text{inst}}(\tau)$ – is called the *instanton*. For the double well potential (1.3), we can perform this integral and get,

$$x_{\text{inst}}(\tau; \tau_0) = a \tanh(\omega(\tau - \tau_0)) \quad (1.28)$$

The solution is sketched below, and is called “an instanton centered at τ_0 ”.



If we flip the sign of τ , we will have a solution that extrapolates from $+a \rightarrow -a$, and we will call this the *anti-instanton*.

The instanton describes a particle that stays at $-a$ for most of the time, and dashes to $+a$ in a very tiny amount of time, and then stays there till eternity. This instantaneous leap is what prompted 'tHooft to name these solutions instantons.

In more quantitative way, the instanton is a well localized object of width $1/\omega$. From the solution (1.28), it is clear that the instanton travels from one minimum to the other in a time of order $1/\omega$.

Now, we can perform the semi-classical approximation around the instanton solution. We have,

$$S[x_{\text{inst}} + \delta x] = S[x_{\text{inst}}] + \int d\tau \frac{1}{2} \delta x(\tau) \left(-\frac{d^2}{d\tau^2} + V''(x_{\text{inst}}) \right) \delta x(\tau) + \dots$$

Using the general result (1.18), we end up with,

$$\langle a | e^{-\hat{H}T/\hbar} | -a \rangle \approx e^{-S_{\text{inst}}/\hbar} \frac{1}{\sqrt{\det\{-\partial_\tau^2 + V''(x_{\text{inst}})\}}} \quad (1.29)$$

There is one issue - this is wrong! The place where we have erred is the assumption that all eigen values are positive. It turns out that there is one zero eigen value.

The Zero Mode

The zero mode is a result of the innocuous integration constant τ_0 in the instanton solution. There is an infinite family of instantons, for each τ_0 . So when we say “expand around the instanton solution”, there is an infinite redundancy of which instanton to choose. This is the source of the zero mode.

We can find the zero mode in a simple way. We know that the instanton satisfy the equation of motion,

$$-\frac{d^2 x_{\text{inst}}}{d\tau^2} + V'(x_{\text{inst}}) = 0$$

All instantons, for each τ_0 satisfy the same equation. Therefore, we can “differentiate the equation of motion” with respect to τ_0 and see that,

$$\left(-\frac{d^2}{d\tau^2} + V''(x_{\text{inst}})\right) \frac{\partial x_{\text{inst}}}{\partial \tau_0} = 0$$

We clearly see that $\frac{\partial x_{\text{inst}}}{\partial \tau_0}$ is a null eigen vector of the differential operator of interest.

Is this the only zero mode? The answer is yes. We can understand this by drawing an analogy with the Schrödinger equation. Our eigen value equation is,

$$\left(-\frac{d^2}{d\tau^2} + V''(x_{\text{inst}})\right) x_n = \lambda_n x_n$$

This looks like the time independent Schrödinger equation in one dimension (with τ playing the role of position x). We know that the lowest energy eigen state of the Schrödinger equation has no nodes, the first excited state has one node and so on. Our zero mode, $x_0 = \frac{\partial x_{\text{inst}}}{\partial \tau_0}$ has no nodes. This is because we know that the instanton is a monotonically increasing function. Therefore, its derivative is guaranteed to be non-zero for all τ_0 . Since the zero mode has no nodes, this must be the lowest energy state. Hence, all the other eigen values of the differential operator will have positive eigen values.

Now we know why the result (1.29) is wrong. The challenge now is dealing with the zero mode. For this, we turn one more time to simple calculus.

An Aside : More on Saddle Point Approximation¹

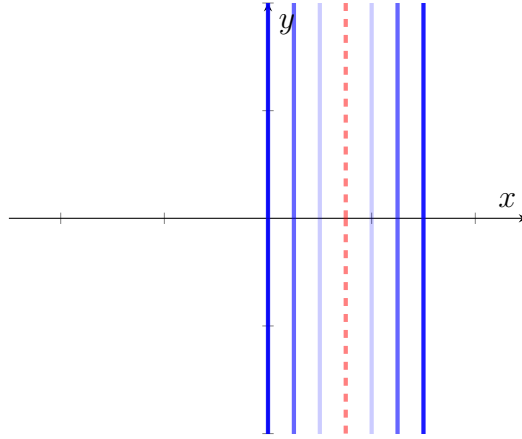
Consider now a two dimensional integral,

$$I = \int_{\mathbb{R} \times \mathbb{R}} dx dy e^{-S(x,y)/\hbar}$$

Suppose there is a well defined minimum of the function $S(x, y)$ at (x_0, y_0) . Then, for small $\hbar$, all the things that we said so far will go through.

Now, we will consider the following: have a well defined minimum in the x -direction, say at x_0 , but have a degeneracy in the y -direction. In pictures, the level curves of $S(x, y)$ is as shown below,

¹I am grateful to Peter Arnold for clearly explaining the subtleties to me!



The red dashed line is at $x = x_0$. $S(x_0, y \in \mathbb{R})$ is the degenerate minima. The “up hill” direction is represented by the increase in opacity of the lines.

Now, we need to modify our saddle point approximation. Since we know that there is a minimum along x , we can expand $S(x, y)$ around x_0 ,

$$S(x, y) = S(x_0, y) + \frac{1}{2}(x - x_0)^2 \frac{\partial^2 S(x_0, y)}{\partial x^2}$$

Then, the integral can be approximated as,

$$I \approx \int_{\mathbb{R}} d(\delta x) \int_{\mathbb{R}} dy \exp \left\{ -\frac{1}{2\hbar} \delta x^2 \frac{\partial^2 S(x_0, y)}{\partial x^2} \right\}$$

The important thing is that the partial derivative $\frac{\partial^2 S(x_0, y)}{\partial x^2}$ is actually independent of y . At x_0 (in our case for any x actually), y is a direction of symmetry. Therefore, our result is actually blowing up, due to the lone integral over y .

If we put on our physicists’ hat, we know that y ultimately is some direction in spacetime. The integral over y is blowing up because space is infinite. All we have to do is to put the entire thing in a box (at least along y). Then, the integral gets approximated to be,

$$I \approx L_y \sqrt{\frac{2\pi\hbar}{S''(x_0)}}$$

We can put some more words to the above result. Since there is a symmetry along y , we can pick a representative y and do the integral for that y . But there are L_y number of different choices to make, so the result simply gets multiplied by L_y .

The integrand that we chose here is a particularly simple one. But the lesson we learned stays true in general. That is, if we have some symmetry of the integrand along some arbitrary (may be even curvilinear) non-compact direction, then the integral will diverge. The way to treat it is to put the theory in a box. This is the kind of arguments that we make when we do the Faddeev-Popov procedure of gauge fixing. What we have done is a baby version of that!

Back to the Zero Mode

We now know why the naive result (1.29) will blow up. We have some symmetry of the integrand (or the action). Let us be meticulous about what it is. The action has time translational symmetry. But the instanton, breaks it due to the presence of τ_0 . This phenomenon is called the *spontaneous breaking of symmetry*. The ground state does not obey some symmetries of the action. Instead, there is a continuous family of ground states, parameterized by τ_0 . Therefore, if we “move along the τ_0 direction”, the action do not change, hence the energy also do not change. This flat direction in the field space is called the Goldstone mode or the zero mode. Integrating naively along this flat direction will result in the integral blowing up.

We know what to do. We just have to put the theory in a box along the τ_0 direction. Before that, we can do the integral along all the other directions other than the zero mode. That is, of all the integrals that we have to do over $c_0, c_1, \dots, c_\infty$, we do every one of them except c_0 . We then end up with,

$$\prod_{n=1}^{\infty} \frac{1}{\sqrt{\lambda_n}} \equiv \frac{1}{\sqrt{\det' \{-\partial_\tau^2 + V''(x_{\text{inst}})\}}}$$

The prime denotes the fact that the zero mode is not included. What is left is an integral over c_0 , which blows up. We can simply put c_0 in a box and call it a day. But, we will make it a point to connect the integral over c_0 to an integral over τ_0 , so we have a physical picture of what is going on.

To relate c_0 to τ_0 , let us recall our approximation procedure. We started of with expanding any path $x(\tau)$ as the instanton plus some small fluctuation. Let us consider the particular case, where the fluctuation is a small change in τ_0 . Then, we have,

$$\begin{aligned} x(\tau) &= x_{\text{inst}}(\tau; \tau_0 + \delta\tau_0) \\ &= x_{\text{inst}}(\tau; \tau_0) + \delta\tau_0 \frac{\partial x_{\text{inst}}}{\partial \tau_0} \end{aligned}$$

We can then compare this with,

$$x(\tau) = x_{\text{inst}}(\tau; \tau_0) + c_0 x_0$$

which is a small fluctuation to the instanton due to the zero mode. We know that the zero mode is,

$$x_0 = N \frac{\partial x_{\text{inst}}}{\partial \tau_0}$$

Therefore, the change in the path due to a change in the instanton position looks identical to the change due to the zero mode. To get the right factors, we have to

find the normalization constant. This is simple,

$$\begin{aligned}
1 &= N^2 \int_{-T/2}^{T/2} d\tau \left(\frac{\partial x_{\text{inst}}}{\partial \tau_0} \right)^2 \\
&= N^2 \int_{-T/2}^{T/2} d\tau \left(-\frac{\partial x_{\text{inst}}}{\partial \tau} \right)^2 \\
&= N^2 \int_{-T/2}^{T/2} d\tau \left[\frac{1}{2} \left(\frac{\partial x_{\text{inst}}}{\partial \tau} \right)^2 + V(x_{\text{inst}}) \right] \\
&= N^2 S_{\text{inst}}
\end{aligned}$$

In the second to third step, we used the energy conservation relation (1.26).

We therefore have,

$$dc_0 = \frac{1}{N} d(\delta\tau_0) = \sqrt{S_{\text{inst}}} d\tau_0$$

Now that we have exchanged the integral over c_0 with τ_0 , we have a natural “box” to put τ_0 in. τ_0 have to be between $-T/2$ and $T/2$. Therefore, we have the result,

$$\begin{aligned}
\langle a | e^{-\hat{H}T/\hbar} | -a \rangle &\approx e^{-S_{\text{inst}}/\hbar} \left(\frac{S_{\text{inst}}}{2\pi\hbar} \right)^{1/2} \int_{-T/2}^{T/2} d\tau_0 \frac{1}{\sqrt{\det' \{-\partial_\tau^2 + V''(x_{\text{inst}})\}}} \\
&= T \left(\frac{S_{\text{inst}}}{2\pi\hbar} \right)^{1/2} e^{-S_{\text{inst}}/\hbar} \frac{1}{\sqrt{\det' \{-\partial_\tau^2 + V''(x_{\text{inst}})\}}} \quad (1.30)
\end{aligned}$$

We have assumed that the determinant is independent of τ_0 . We are not wrong because the determinant is almost always a constant (and equal to the oscillator result), except for a small width of order $1/\omega$. But the range of integration T is much much larger than this width.

We have a much clearer picture of what is going on. To evaluate the matrix element $\langle a | e^{-\hat{H}T/\hbar} | -a \rangle$ semi-classically, we should expand the path integral around an instanton. There are infinitely many instantons, parameterized by τ_0 . Since the action do not depend on τ_0 , all instantons are equally good. Therefore, we can do the integral by picking one instanton, and considering small fluctuations around it. The fluctuation along the zero mode (i.e, the $c_0 x_0$ part) actually corresponds to translating the instanton. Therefore, we do not have to do the integral along c_0 if we want to use one reference instantons out of infinitely many. We then simply perform all the integrals except c_0 . This result should be multiplied by the number of choices of instantons, which is a continuous infinity of choices corresponding to $\tau_0 \in (-T/2, T/2)$.

There is a further re-writing that is done. The motivation is that $V''(x_{\text{inst}})$ is almost always ω^2 , except for a small amount of time of the order $1/\omega$, when the instanton

makes its move. Therefore, it is convenient (later to extract the large time behavior) to write,

$$\frac{1}{\sqrt{\det' \{-\partial_\tau^2 + V''(x_{\text{inst}})\}}} = \sqrt{\frac{\det \{-\partial_\tau^2 + \omega^2\}}{\det' \{-\partial_\tau^2 + V''(x_{\text{inst}})\}}} \frac{1}{\sqrt{\det \{-\partial_\tau^2 + \omega^2\}}}$$

Denote,

$$K = \sqrt{\frac{S_{\text{inst}}}{2\pi\hbar}} \sqrt{\frac{\det \{-\partial_\tau^2 + \omega^2\}}{\det' \{-\partial_\tau^2 + V''(x_{\text{inst}})\}}} \quad (1.31)$$

We can use the result of the harmonic oscillator, that,

$$\frac{1}{\sqrt{\det \{-\partial_\tau^2 + \omega^2\}}} \xrightarrow{T \rightarrow \infty} e^{-\omega T/2}$$

to have our final result,

$$\langle a | e^{-\hat{H}T/\hbar} | -a \rangle \approx T e^{-\omega T/2} K e^{-S_{\text{inst}}/\hbar} \quad (1.32)$$

1.6 The Instanton Gas

We are done with both the computations. So we should be able to find the splitting energy now, right? Well, no. It turns out there are more (actually infinitely many) approximate solutions that are equally good starting points for a semi-classical approximations. These are strings of instantons and anti-instantons, which is called the *instanton gas*. This is what we will discuss now.

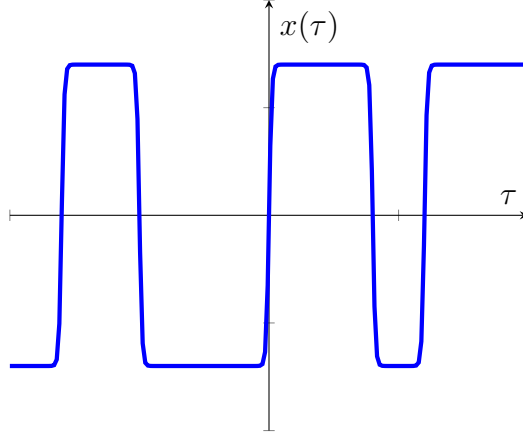
As we saw already, the instanton is a well localized object, of width of order $1/\omega$. Therefore, if we take an instanton, an anti-instanton and another instanton, one followed by the next with long time intervals in between, we will have an approximate solution to the equations of motion. In general, we can take a string of instantons and anti-instantons, with separation between each of them $L \gg 1/\omega$, we will have an approximate solution. Let us try to understand this better.

For simplicity, let us take an instanton, followed by an anti-instanton. Then, the approximate solution that we propose is,

$$x(\tau) = x_{\text{inst}}(\tau; 0) \times x_{\text{inst}}(-\tau; L)$$

The assumption here is $L \gg 1/\omega$. As long as this condition is true, we are guaranteed that the equation of motion will be satisfied approximately. The correction will be (inferring from the exponential nature of the instanton) of order $\mathcal{O}(e^{-\omega L})$.

The action for a string of n instantons is approximately nS_{inst} , and the correction is of the order $\mathcal{O}(e^{-\omega L})$. Therefore, for semi-classical approximation, a string of instantons is also a good background about which we can expand. This is known as the *dilute gas approximation*. An example of a dilute gas is shown below,



Let us now perform the dilute gas approximation. We take a string of n instantons, at times,

$$-\frac{T}{2} < \tau_1 < \tau_2 < \cdots < \tau_n < \frac{T}{2}$$

We will assume that,

$$\tau_{k+1} - \tau_k \gg 1/\omega$$

The action for this string of instantons is,

$$S_{n - \text{inst}} \approx n S_{\text{inst}}$$

We can adapt the calculation we did for one instanton (1.30) to this case. Now, we have n zero modes, for each of the (anti-)instanton. The determinant without zero modes is,

$$\frac{1}{\sqrt{\det' \{-\partial_\tau^2 + V''(x_{n-\text{inst}})\}}} \quad (1.33)$$

where $x_{n-\text{inst}}$ denotes the n instanton approximate solution. We should have n to be odd so that the particle starts at one minimum and ends at the other. Therefore, we have the result,

$$\langle a | e^{-\hat{H}T/\hbar} | -a \rangle \approx \sum_{\text{odd } n} e^{-\frac{n S_{\text{inst}}}{\hbar}} \left(\frac{S_{\text{inst}}}{2\pi\hbar} \right)^{\frac{n}{2}} \int_{-T/2}^{T/2} d\tau_1 \int_{\tau_1}^{T/2} d\tau_2 \cdots \int_{\tau_{n-1}}^{T/2} d\tau_n \frac{1}{\sqrt{\det' \{-\partial_\tau^2 + V''(x_{n-\text{inst}})\}}}$$

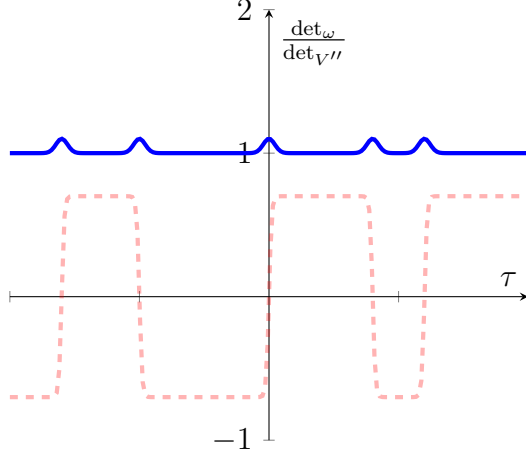
The determinant is almost always constant, and equal to the harmonic oscillator result, except in the vicinity of the instanton. But this width, as per our assumption is very small compared to the range of integration, so we can assume that the determinant terms are independent of the instanton positions. Then, the integrals evaluate to,

$$\int_{-T/2}^{T/2} d\tau_1 \int_{\tau_1}^{T/2} d\tau_2 \cdots \int_{\tau_{n-1}}^{T/2} d\tau_n = \frac{T^n}{n!}$$

What is left is some massaging of (1.33). We multiply and divide by the harmonic oscillator result to get,

$$\frac{\sqrt{\det\{-\partial_\tau^2 + \omega^2\}}}{\sqrt{\det'\{-\partial_\tau^2 + V''(x_{\text{n-inst}})\}}} \frac{1}{\sqrt{\det\{-\partial_\tau^2 + \omega^2\}}}$$

The second factor $\frac{1}{\sqrt{\det}}$ will give us the large time behavior $e^{-\omega T/2}$ as before. The first factor is almost always unity, except around the vicinity of the (anti-)instanton. Pictorially, this factor looks like the blue curve below,



The dashed red line is the instanton gas. The full blue curve is a product of each of the “bumps”². That is,

$$\frac{\sqrt{\det\{-\partial_\tau^2 + \omega^2\}}}{\sqrt{\det'\{-\partial_\tau^2 + V''(x_{\text{n-inst}})\}}} = \prod_j \frac{\sqrt{\det\{-\partial_\tau^2 + \omega^2\}}}{\sqrt{\det'\{-\partial_\tau^2 + V''(x_{\text{inst}}^{(j)})\}}}$$

where $x_{\text{inst}}^{(j)}$ denotes the j^{th} instanton. Now, each of the factor on the right-hand side is defined in the one instanton case (1.31),

$$\frac{\sqrt{\det\{-\partial_\tau^2 + \omega^2\}}}{\sqrt{\det'\{-\partial_\tau^2 + V''(x_{\text{inst}}^{(j)})\}}} = \sqrt{\frac{2\pi\hbar}{S_{\text{inst}}}} K$$

Using this, we have,

$$\langle a | e^{-\hat{H}T/\hbar} | -a \rangle \approx e^{-\omega T/2} \sum_{\text{odd } n} \frac{1}{n!} (KT e^{-S_{\text{inst}}/\hbar})^n = e^{-\omega T/2} \sinh(KT e^{-S_{\text{inst}}/\hbar}) \quad (1.34)$$

Likewise, the semi-classical transition amplitude to go from a to a is also found by a dilute gas approximation, but this time n is even (including zero). This gives us,

$$\langle a | e^{-\hat{H}T/\hbar} | a \rangle \approx e^{-\omega T/2} \sum_{\text{even } n} \frac{1}{n!} (KT e^{-S_{\text{inst}}/\hbar})^n = e^{-\omega T/2} \cosh(KT e^{-S_{\text{inst}}/\hbar}) \quad (1.35)$$

²Unsurprisingly, the way I made the blue curve is to plot $\prod_j (1 + e^{-(x-x_j)^2})$.

We can check if these results are compatible with our assumption that the instantons are far apart. To this end, recall that the sum $\sum_n \frac{x^n}{n!}$ for some x will increase until n is of the order x , and after that starts decreasing rapidly. In our case, the sum contributes significantly when,

$$n \lesssim KT e^{-S_{\text{inst}}/\hbar}$$

Or in other words, the sum contributes significantly only when the density of the instantons n/T is exponentially small, which sits well with our assumption.

Energy Splitting

Now, we are ready to compute the energy splitting between the first and excited states. We have,

$$\langle \text{ground} | e^{-\hat{H}T/\hbar} | \text{ground} \rangle = 2 \left(\langle a | e^{-\hat{H}T/\hbar} | a \rangle + \langle a | e^{-\hat{H}T/\hbar} | -a \rangle \right)$$

and,

$$\langle \text{first} | e^{-\hat{H}T/\hbar} | \text{first} \rangle = 2 \left(\langle a | e^{-\hat{H}T/\hbar} | a \rangle - \langle a | e^{-\hat{H}T/\hbar} | -a \rangle \right)$$

Using our dilute gas results,

$$\langle \text{ground} | e^{-\hat{H}T/\hbar} | \text{ground} \rangle \xrightarrow{T \rightarrow \infty} 2 \exp \left\{ - \left(\frac{\omega}{2} - K e^{-S_{\text{inst}}/\hbar} \right) T \right\}$$

and,

$$\langle \text{first} | e^{-\hat{H}T/\hbar} | \text{first} \rangle \xrightarrow{T \rightarrow \infty} 2 \exp \left\{ - \left(\frac{\omega}{2} + K e^{-S_{\text{inst}}/\hbar} \right) T \right\}$$

Therefore, we have,

$$E_0 = \hbar \left(\frac{\omega}{2} - K e^{-S_{\text{inst}}/\hbar} \right) \quad \text{and} \quad E_1 = \hbar \left(\frac{\omega}{2} + K e^{-S_{\text{inst}}/\hbar} \right) \quad (1.36)$$

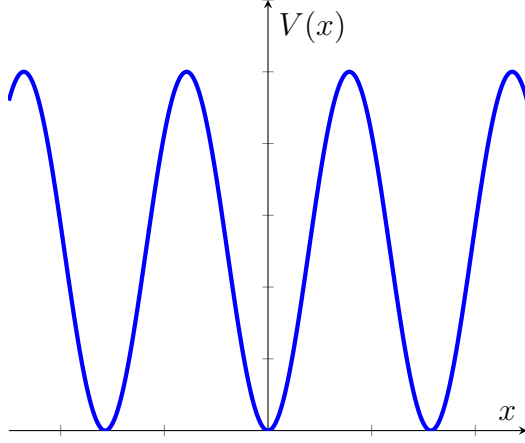
Strictly speaking, the above formula should not be trusted. We have not calculated the usual perturbative corrections to the eigen states due to the potential itself. These are supposed to come in of the order $\mathcal{O}(\hbar^2)$. But due to the symmetry of the potential, we expect the perturbative corrections to be the same for all eigen states. Therefore, the splitting energy can be found and can be trusted. It is given by,

$$\Delta E = E_1 - E_0 = 2K e^{-S_{\text{inst}}/\hbar} \quad (1.37)$$

The split is exponentially small!

1.7 Periodic Potentials and Bloch Waves

Let us consider a periodic potential, like one shown below,



Our classical expectation is that the ground state is localized at each minimum of the potential, with an infinite degeneracy. We know that this is incorrect, and the true ground states are the so-called *Bloch waves*, which are the eigen states of the translation operator along x . We will arrive at the same result, but through instantons.

In the inverted version of the potential above, we can already see that the instantons are the same as what we had. The novelty is that the instantons can now start at any position $x = ja$ where it's a minimum, and end at the next minimum given by $j + 1$. Anti-instantons will start from j and end at $j - 1$.

We can therefore have a dilute gas of instantons as before. We do not have to restrict the instantons and anti-instantons to come in alternating pairs anymore. Furthermore, we can have them in any order, as long as the next instanton starts where the previous instanton ended. Also, the total number of instantons minus the number of anti-instantons must equal the change $\Delta j = \frac{\Delta x}{a}$. Hence, we have the dilute gas sum,

$$\langle j_- | e^{-\hat{H}T/\hbar} | j_+ \rangle \underset{T \rightarrow \infty}{\approx} e^{-\omega T/2} \sum_{n=1}^{\infty} \sum_{n'=1}^{\infty} \frac{(KT e^{-S_{\text{inst}}/\hbar})^{n+n'}}{n!n'!} \delta_{n-n', j_+ - j_-} \quad (1.38)$$

We re-write the Kronecker delta using,

$$\delta_{ab} = \int_0^{2\pi} \frac{d\theta}{2\pi} e^{i(a-b)\theta}$$

Then the sums are easy, and we end up with,

$$\langle j_- | e^{-\hat{H}T/\hbar} | j_+ \rangle \underset{T \rightarrow \infty}{\approx} e^{-\omega T/2} \int_0^{2\pi} \frac{d\theta}{2\pi} e^{i(j_- - j_+)\theta} \exp\{2KT \cos \theta e^{-S_{\text{inst}}/\hbar}\} \quad (1.39)$$

We see that there is a continuous set of energies, each labelled by the angle θ ,

$$E(\theta) = \frac{1}{2}\hbar\omega + 2\hbar K \cos \theta e^{-S_{\text{inst}}/\hbar} \quad (1.40)$$

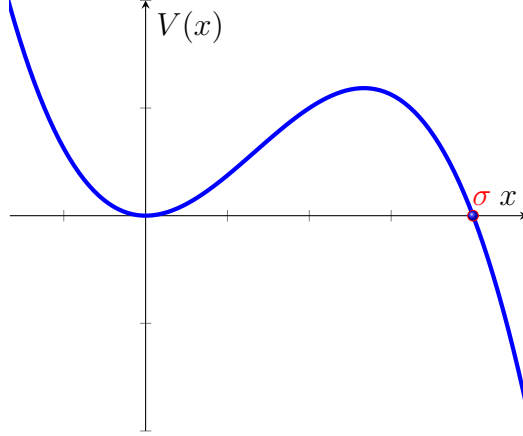
The energy eigen states are also easy to read off from here, as we saw from (1.8),

$$\langle \theta | j \rangle \propto e^{ij\theta} \quad (1.41)$$

These results are the correct answers!

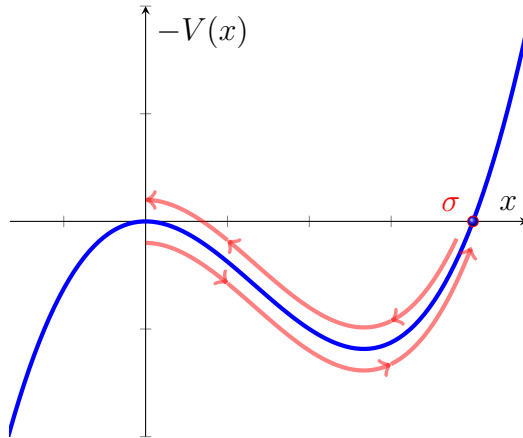
1.8 Unstable States and Bounces

So far, we had the double well potential which is bounded. We will now consider an unbounded potential of the shape,



Our classical (or semi-classical) intuition would tell us that there is a state that is localized at the minimum, which we take to be at $x = 0$. But we know better. This state, as a result of tunneling will be unstable. We will now try to explain this using the same methods.

Let us just do what we have done so far. We flip the potential upside down, and think about a classical mechanics problem of a particle starting at the top of the hill at $x = 0$ and going to $x = \sigma$ and then coming back to the top of the hill in infinite time. This trajectory will look like,



This trajectory will be henceforth called the *bounce*.

Naive Calculation

Let us be naive and repeat the instanton gas procedure. To find the matrix element $\langle x = 0 | e^{-\hat{H}T/\hbar} | x = 0 \rangle$, we take n widely separated bounces as the background around the expansion and sum over all possible n . The answer will be the same in form with appropriate re-definitions of S_{cl} and ω . We can sum over all n , and don't have to restrict to odd or even numbers. This gives us,

$$\langle x = 0 | e^{-\hat{H}T/\hbar} | x = 0 \rangle \approx e^{-\omega T/2} \exp\{KT e^{-S_0/\hbar}\} \quad (1.42)$$

This naively seems to give us the correction to the energy of the “ground state”,

$$E_0 = \frac{1}{2}\hbar\omega + \hbar K e^{-S_0/\hbar} \quad (1.43)$$

There seems to be three issues with the result. Let us list them here,

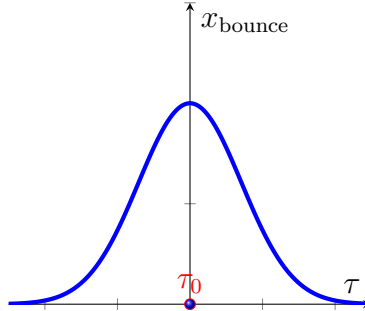
1. There is supposed to be no ground state. The state localized at the minimum is rendered unstable due to tunneling.
2. We have no right in retaining an exponential correction in $\hbar$, since we have not computed the usual power series perturbative corrections.
3. K is imaginary.

The third point needs some explaining. Recall that K is,

$$K = \sqrt{\frac{S_0}{2\pi\hbar}} \sqrt{\frac{\det\{-\partial_\tau^2 + \omega^2\}}{\det'\{-\partial_\tau^2 + V''(x_{\text{bounce}})\}}}$$

With the double well potential instanton, we saw that the lowest eigen value of the operator in the denominator was zero, and the prime denotes that zero eigen value is omitted. The zero eigen value was due to instanton breaking the time translation symmetry.

It is clear that the bounce will also break time translational symmetry, hence will correspond to a zero mode. This mode is omitted from the determinant and repackaged into the factor of T . But the new issue is that in the case of the bounce, the zero mode is not the lowest eigen value state. To see this, we just have to make a plot of the bounce. The bounce will look like,



The zero mode is proportional to $\frac{\partial x_{\text{bon}}}{\partial \tau_0}$, and it will have one node, at τ_0 . Once again using the intuition of one dimensional Schrödinger equation, we are sure that there is a nodeless eigen function – call it x_{-1} – which has an eigen value smaller than zero. Hence, K involves the square root of a negative number, hence imaginary.

A Change of Perspective Saves the Day!

It turns out that the naive calculation we have done is almost correct, up to a factor of two! All we have to do is to interpret the result correctly. Error number 1 is remedied by error 3 itself. Since the correction to the energy is imaginary, the corresponding state will decay as time evolves! The decay width is,

$$\text{Im}\{E_0\} = \frac{\Gamma}{2} = \hbar|K| e^{-S_0/\hbar} \quad (1.44)$$

If we use the above result, then error 2 is also remedied. The leading contribution to the decay width is what we have. There are no $\mathcal{O}(\hbar^2)$ perturbative corrections to the decay width. This is because perturbation theory assumes that we are very close to the minimum at $x = 0$, as opposed to the bounce solution which is “spread out” in space. This guarantees that all perturbative corrections will be on the real part of E_0 .

A Missing Factor of Two

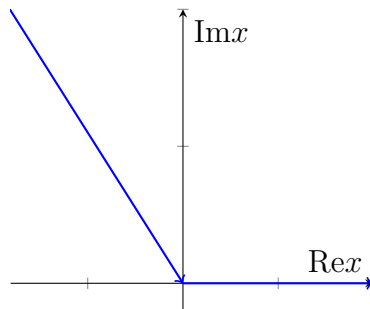
Although qualitatively correct, (1.44) misses a factor of two. Finding this factor is not of extreme importance to understand the basics of instanton methods. We will now find this factor for completeness. The correct answer for the decay rate is,

$$\Gamma = \hbar|K| e^{-S_0/\hbar} \quad (1.45)$$

The origin of this factor is complex analysis. To get the gist, let us consider an ordinary integral,

$$I = \int_{\mathbb{R}} dx \exp\{-x^2 - x^3\}$$

The integral diverges for $x \rightarrow -\infty$. We should somehow change the contour from $\mathbb{R}$ to something else so that the integral is convergent. A contour of the form shown below will do the trick,



Then, we can do the steepest descent approximation around $x = 0$, and the imaginary part of the integral will be,

$$\text{Im}\{I\} \approx \text{Im}\left\{\int_{i\infty}^0 dx \exp\{-x^2 \cdot \text{constants}\}\right\}$$

The imaginary part comes from only a half of the total integration region. The $(0, \infty)$ region has no contribution to the imaginary part. Therefore, the imaginary part will have a factor of half.

The potential we have is unbounded only on one side. Therefore, to do the path integral in the negative eigen value direction, we will have to deform the contour just like we did now for the ordinary integral. Since this deformation is needed only on one side, we will have a factor of half when we perform the steepest descent method for the negative eigen value direction. Hence, we have justified the result for the decay rate given by (1.45).

References